

Isoliner · Topography: from DEM to catchments

A cheat sheet for the "2. Topography" group · QGIS 3.16+ · Isoliner plugin v4 · pure NumPy, no GRASS or SAGA

Pipeline: from a DEM to Topo2Raster

Step 1. DEM - 2.01 Download DEM by extent. Set the extent, keep the defaults. Output: a 30 m terrain raster in a metric CRS.

Step 2. Contours - 1.04 Isolines from raster. Input: the DEM of step 1. A 10 m contour step, the **ELEV** field, do not raise the minimum length (short closed lines are hilltops), turn contour polygons off.

Step 3. Streamlines - 2.06 River network. Input: the DEM of step 1. Threshold 1000 cells, a denser network - a lower threshold. Line vertices already run downstream, as Topo2Raster requires.

Step 4. Extreme marks - 2.09 Peaks and pits. Input: the DEM of step 1. Radius 500 m, relief 20 m. Output: points with the **z**, **drop** and **kind** (peak or pit) fields, the elevation right in the geometry Z - the cure for the flat caps at closed contours, both tops and bottoms.

Step 5. Water bodies and cliffs - 2.02 Download base topography by extent. The same extent. From OSM we take what a raster cannot give: water bodies (flat planes) and cliffs (barriers). OSM watercourses and peaks are a fallback for steps 3-4 when ele is filled.

Step 6. Terrain - 2.03 Topo2Raster. The contours of step 2 (the ELEV field), streamlines - the network of step 3, elevation points - the peaks of step 4 (the z field), lakes - the water bodies of step 5, cliffs as barriers (the step is not smeared). Cell 30 m, the extent empty. Output: the topographic terrain raster.

Control: 1.04 over the result with the same step, overlaid on the contours of step 2 - the lines must coincide. Contours from a digitized plan replace steps 1-2. All outputs land in the **Topography** group of the layer tree.

Tool reference: input, output, key parameters

Tool	Input	Output	Key parameters
2.01 Download DEM	map extent	GeoTIFF float32, meters	source: GLO-30 (DSM) or GEDTM30 (DTM, forest removed). Cell 30 m. CRS: empty = project/UTM
2.02 OSM base map	map extent	watercourses, water bodies, peaks (ele), cliffs	coastline off. Large extent: shrink it or raise the limit
1.04 Isolines from raster	DEM	lines with ELEV	step 5-10 m. Do not raise min length: short closed lines are hilltops. Moderate smoothing. Polygons off
2.03 Topo2Raster	contours (elevation field), points, streamlines, cliffs, lakes	GeoTIFF float32	points or contours required. Streamlines run downstream: OSM and 2.06 fit as is. Edge: node Z (river slope) > field elevation (plane) > shore minimum
2.04 Terrain preparation	DEM	prepared DEM	FPDEMS smoothing (edge-preserving) and/or depression filling (epsilon 0.001 for D8). Both by checkboxes
2.05 Flow D8	DEM	directions (ArcGIS codes, byte) + accumulation (cells, float32)	codes: E=1, SE=2, S=4, SW=8, W=16, NW=32, N=64, NE=128, sink=0
2.06 River network	DEM	lines: order (Strahler), acc_out, length_m	threshold = head catchment / cell area. 1000 at 30 m $\approx$ 0.9 km ²
2.07 Basins	DEM, pour points (opt.)	polygons: basin, area_m2 (+ label raster)	without points: from mouths by threshold. Point snap to the accumulation max, 150 m
2.08 Slope and aspect	DEM	two rasters, degrees	Horn 3×3 as gdaldem. Aspect: downslope azimuth, flats = -1
2.09 Peaks and pits	DEM	points: z, drop, kind (PointZ)	both signs always. A 500 m radius suppresses secondary tops, a 20 m relief cuts bumps and puddles
2.15 Gauge report	DEM, gauge points	points with catchment and slope	snapping to the accumulation maximum. HTML report
2.16 Catchment. Lines and outlines	DEM, lines or polygons	catchment polygon, area	a polygon is an intake as a whole, both the outline and the area inside. For a pit that removes the dependence on flow directions in the hollow
2.18 Cut and fill	two surfaces, or a surface and an elevation	difference raster, HTML statement	the sign is "after minus before". Grids are aligned bilinearly. The dead band cuts background noise

Common pitfalls

- **A degree CRS.** The group works in meters. 2.01 reprojects itself: an empty target CRS = the project CRS if metric, otherwise UTM at the center. User CRSs without an EPSG code are fully supported.
- **The river threshold is tuned.** Start at 1000 and adjust threefold. A DEM check: overlay the 2.06 network on the OSM watercourses.
- **Overpass is not elastic.** Large requests get cut by OSM servers, on a failure the tool switches to a mirror. Keep the extent modest.
- **Closed basins.** Karst and subsidence troughs are erased by the hydro correction: uncheck the filling.

{Data: Copernicus DEM © ESA · GEDTM30 © OpenGeoHub CC BY 4.0 · © OpenStreetMap, ODbL. Plugin: plugins.qgis.org/plugins/grid_isolines/ · The manual ships with the plugin (doc/Isoliner_en.pdf) · Inform++ LLC · www.informpp.ru}